

Sri Lanka Institute of Information Technology



Software Engineering | SE2030

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Group ID – 55 (2025-Y2-S1-MLB-B3G2-05)

LEARN - MATE

Web-based Educational Institute Management System

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Contents

Introduction.....	3
Sprint summaries.....	4
Use Case Diagram	6
Class Diagram.....	7
Activity Diagrams	8
Ethical Considerations	14

Introduction

In today's educational landscape, efficient management of academic and administrative processes is crucial for success. Many institutes, however, still rely on manual, paper-based systems and disconnected digital records like spreadsheets. This leads to significant challenges including data fragmentation, operational inefficiency, communication gaps, and a lack of real-time information for students and parents.

LEARN MATE is designed to directly address these issues. It is a comprehensive, web-based management system that serves as a centralized platform to automate and streamline the core functions of an educational institute. By replacing outdated methods with an integrated solution, LEARN MATE aims to create a more efficient, transparent, and connected environment for all stakeholders.

Our vision is to develop a single, intuitive application that connects the entire institute community. The core objectives of LEARN MATE are:

- ✓ For Management: To provide a high-level dashboard for monitoring key performance indicators like attendance, academic results, and fee status.
- ✓ For Administrative Staff: To automate routine tasks such as student enrollment, timetable management, and fee tracking, thereby reducing manual effort and errors.
- ✓ For Teachers: To offer a dedicated platform for uploading materials, recording attendance and marks, and communicating with students and parents.
- ✓ For Students & Parents: To create a transparent portal for accessing timetables, viewing results, paying fees online, and receiving important announcements.
- ✓ For Student Support: To enable early intervention by alerting support officers to students with low attendance or poor academic performance.

The LEARN MATE system is built around six major functional modules:

1. User Management: Secure, role-based access control for all users.
2. Class & Timetable Management: Creation and publication of personalized schedules.
3. Attendance Tracking: Digital recording and real-time monitoring of student attendance.
4. Exams & Marks Management: Centralized input and transparent publication of academic results.
5. Fee Management: Online fee payment processing and automated payment reminders.
6. Communication & Notifications: A central hub for announcements and automated alerts.

LEARN MATE is being developed as a robust and user-friendly web application. The technology stack, as per module requirements, includes Java for the backend logic and MySQL for the database. The system will feature a responsive design, ensuring a seamless experience on desktops, tablets, and mobile devices. Key non-functional requirements such as performance, security, and usability are central to our development process.

The LEARN MATE project represents a significant step towards modernizing the operational framework of educational institutes. Our team has a clear development plan and is committed to delivering a functional, high-quality system within the semester timeline. This system will not only solve immediate operational challenges but also lay a strong digital foundation for the institute's future growth.

Sprint summaries

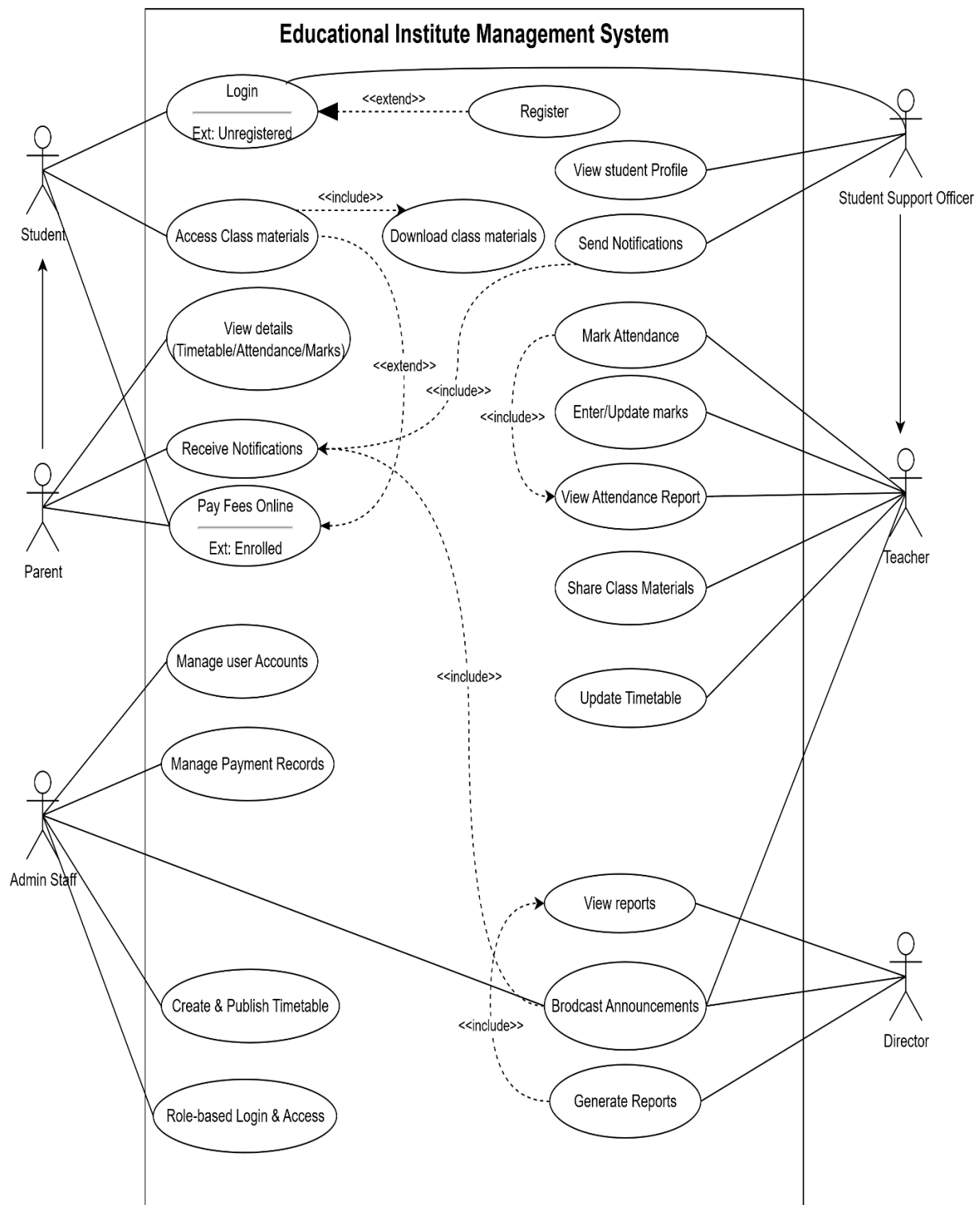
Sprint Plan 1

Task ID	User Story (Summary)	Priority	Assigned Member	Status
US01	Manage user accounts (CRUD)	High	IT24100300	Complete
US02	Create and publish timetables	High	IT24100237	Complete
US03	Mark student attendance	High	IT24100263	Complete
US04	Enter and manage marks	High	IT24100314	Complete
US05	View personalized timetable	High	IT24100237	Complete
US06	Access grades and marks	High	IT24100314	Complete
US07	View child's academic results	Medium	IT24100314	Not Complete
US08	Track child's attendance	Medium	IT24100263	Not Complete
US09	Manage fee records	High	IT24100263	Not Complete
US10	View fee balances and pay online	High	IT24100283	Not Complete
US14	View own attendance history	Medium	IT24100263	Complete
US15	View daily schedule	Medium	IT24100237	Complete

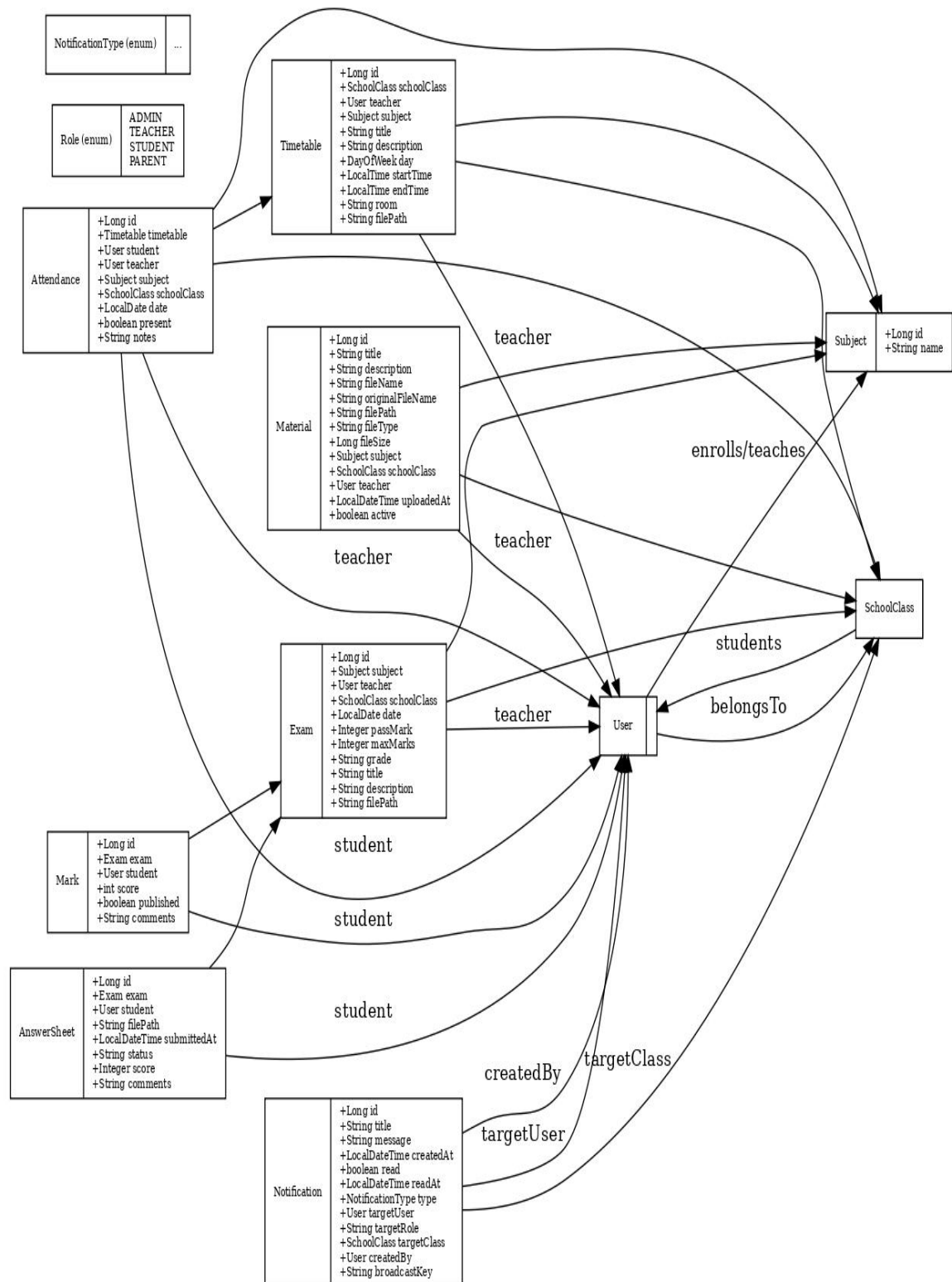
Sprint Plan 2

Task ID	User Story (Summary)	Priority	Assigned Member	Status
US11	View real-time dashboard	Medium	IT24100300	Complete
US12	Broadcast announcements	High	IT24100239	Complete
US13	Receive notifications	High	IT24100239	Complete
US16	Send messages to parents	High	IT24100239	Not Complete
US19	Generate summary reports	Medium	IT24100300	Not Complete
US20	Receive school announcements	High	IT24100239	Complete
US17	Get alerts for low attendance	Medium	IT24100263	Not Complete
US18	Get alerts for failing grades	Medium	IT24100314	Not Complete
US21	Access student's complete profile	Medium	IT24100300	Not Complete
US22	View historical academic data	Medium	IT24100314	Complete
US23	Send private messages	Medium	IT24100239	Not Complete
US24	View teacher activities overview	Medium	IT24100300	Complete

Use Case Diagram

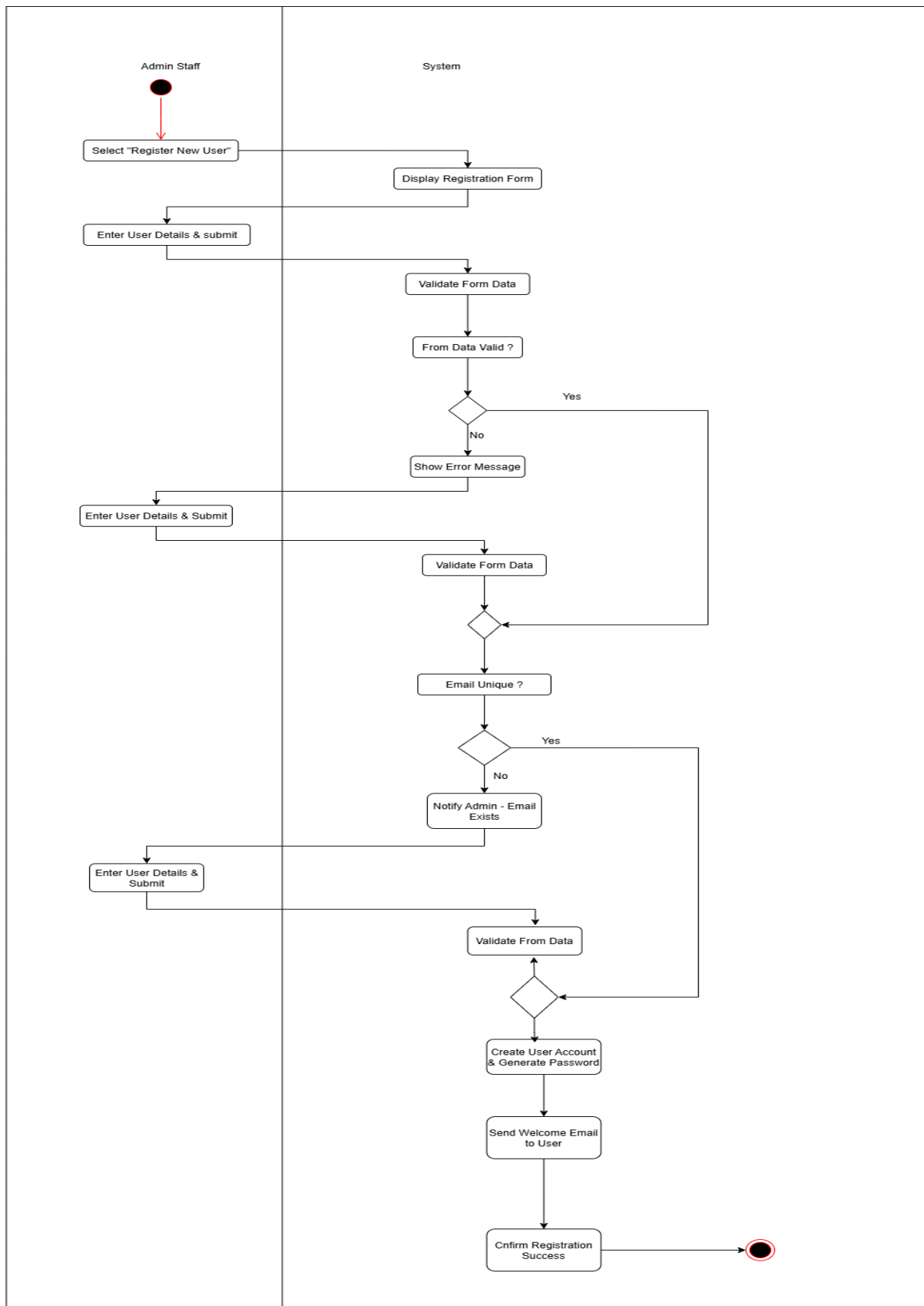


Class Diagram

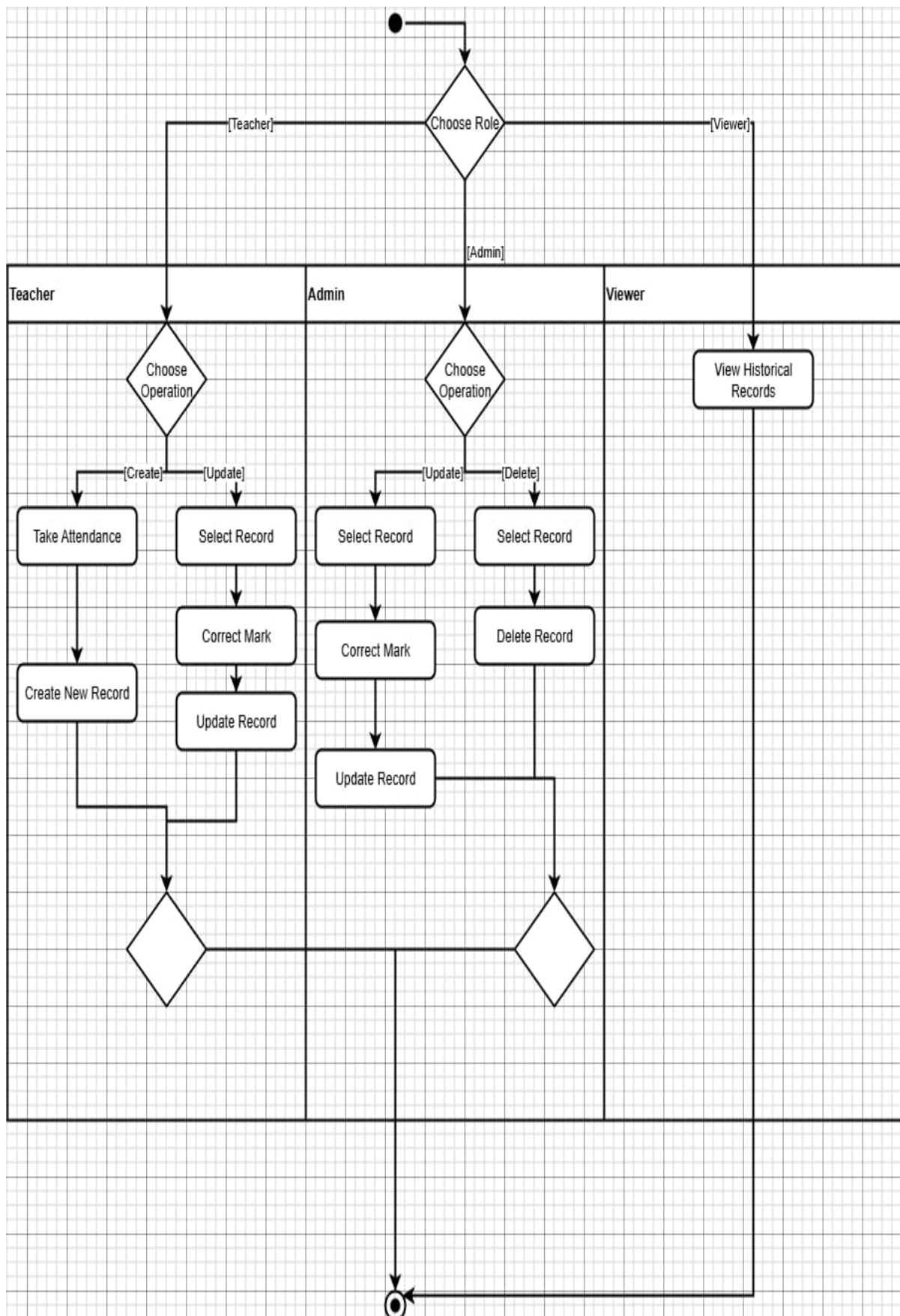


Activity Diagrams

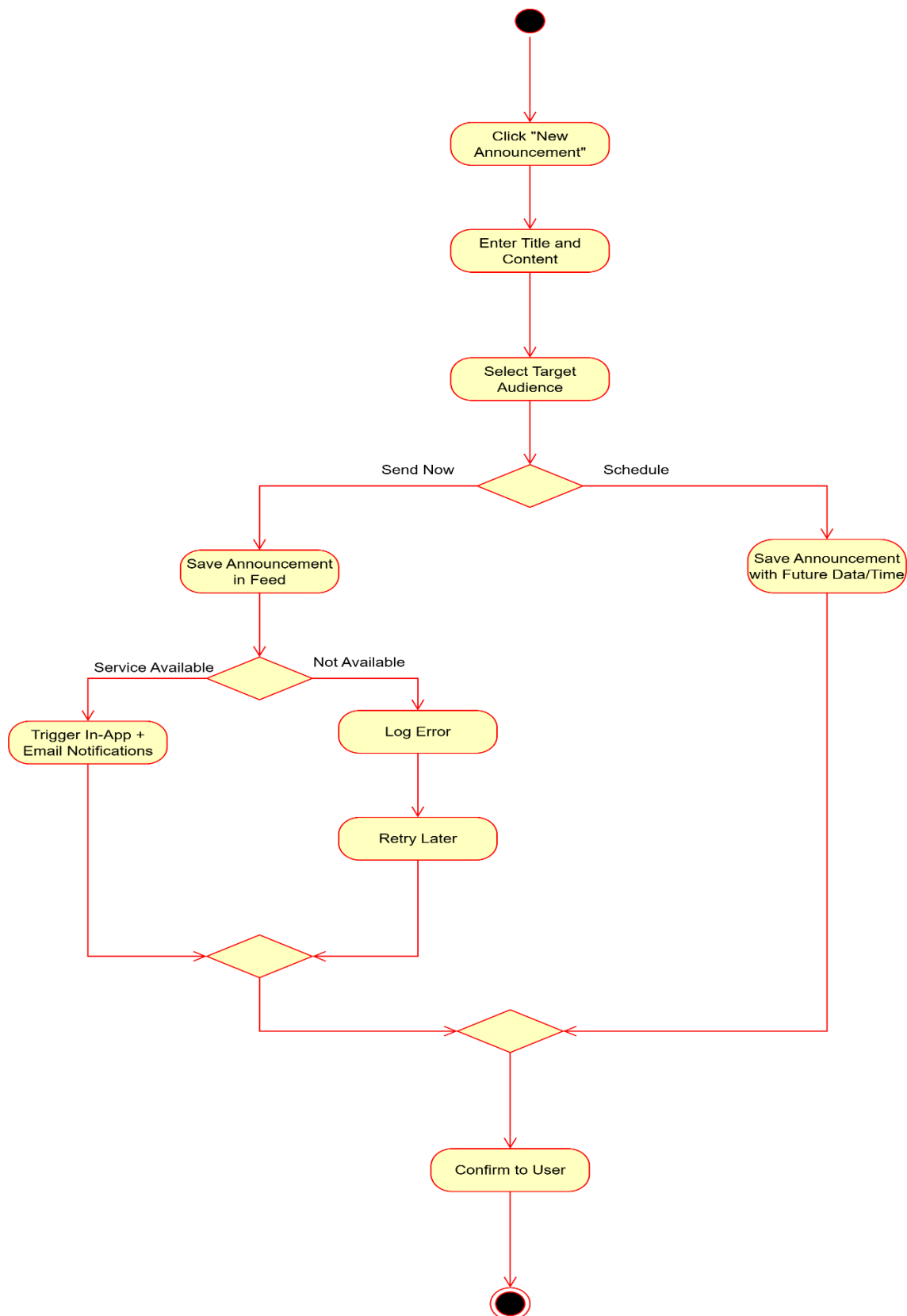
User Management (IT24100300 – Dulakshika A.L.S.H)

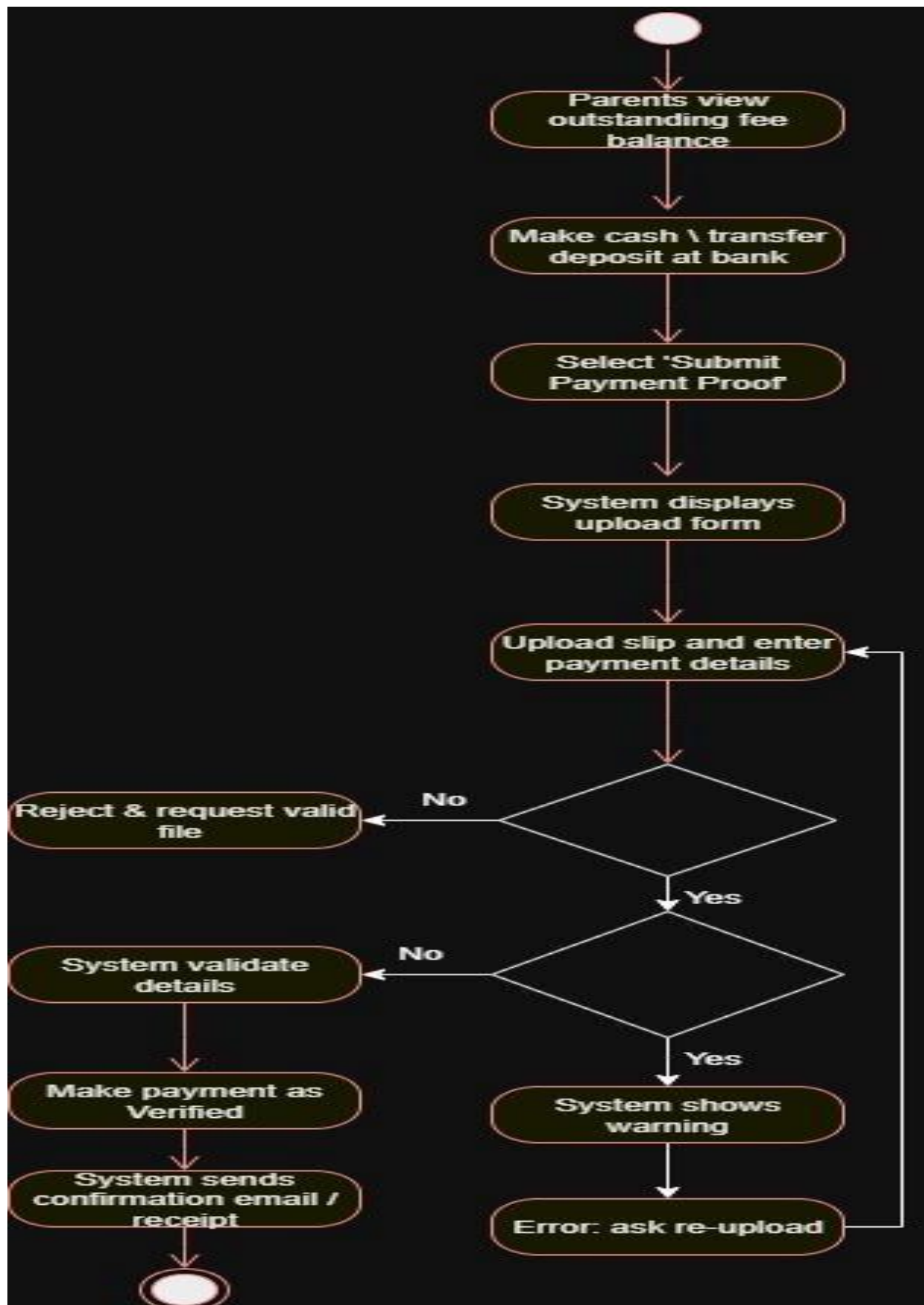


Attendance Tracking (IT24100263 - Dilhani W. P. K. A)

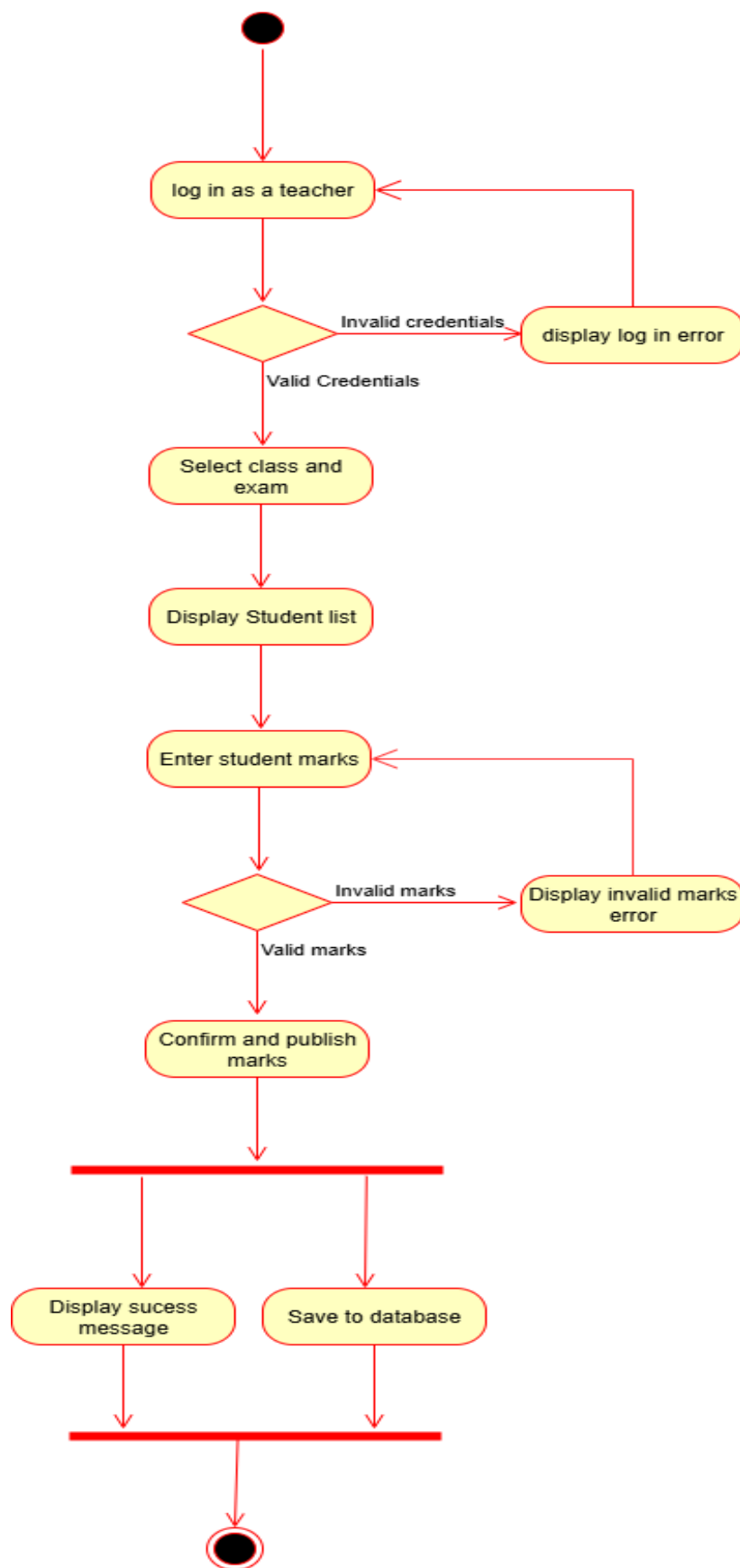


Communication and Notifications (IT24100239 –Disanyaka D.M.C.N)

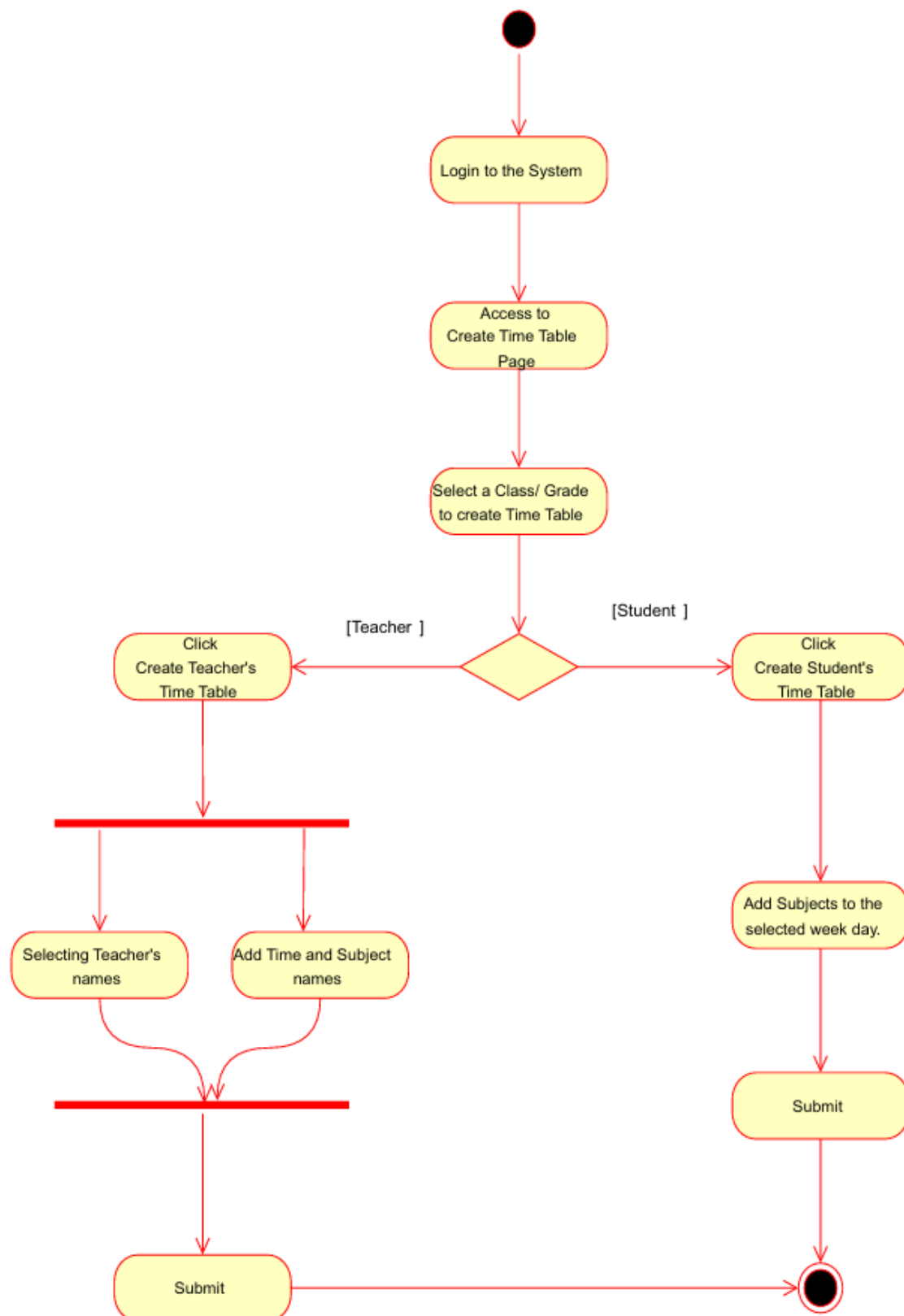




Exams & Marks management (IT24100314 - Dahanayake L.K)



Class & Timetable Management (IT24100237 - Bandara R.M.G.L)



Ethical Considerations

1. Data Privacy and Protection

1.1 Password Security Implementation

The "Learn Mate" system incorporates robust password security measures to protect user accounts and sensitive information:

- **Password Complexity Requirements:** All user passwords must contain a minimum of 8 characters, including: At least one uppercase letter, At least one lowercase letter, At least one special symbol or number, This ensures strong password strength against brute-force attacks
- **Password Encryption:** All user passwords are encoded using industry-standard hashing algorithms (BCrypt) before storage in the database. This ensures that even in the event of a data breach, plain-text passwords remain protected.

1.2 Personal Data Protection

- **Data Minimization:** The system collects only essential personal information required for educational purposes
- **Confidentiality:** Student records, academic performance, and personal information are accessible only to authorized personnel based on role-based permissions
- **Secure Data Transmission:** All data transmitted between the user's browser and the server is encrypted using HTTPS/TLS protocols

2. Ethical Data Usage

2.1 Academic Data Ethics

- **Fair Assessment:** The system ensures unbiased recording and calculation of student marks and grades
- **Transparent Evaluation:** Students and parents can access and review academic results with clear explanation of grading criteria
- **Early Intervention:** Attendance and performance alerts are designed to support student success rather than punitive measures

2.2 Communication Ethics

- **Respectful Communication:** The notification system includes safeguards against spam and inappropriate content
- **Cultural Sensitivity:** Communication templates and content consider diverse cultural backgrounds of users
- **Emergency Protocols:** Clear guidelines for appropriate use of emergency notification features

3. Security and Accountability

3.1 Access Control

- **Role-Based Permissions:** Strict access controls ensure users can only access information relevant to their role
- **Audit Trails:** System maintains logs of significant actions (mark changes, attendance modifications, etc.) for accountability
- **Session Management:** Automatic logout after periods of inactivity to prevent unauthorized access

4. Educational Equity

- The system is designed to provide equal access to educational resources and information for all students, regardless of their background or technical capabilities
- Features like automated notifications help ensure all stakeholders receive important information in a timely manner

5. Compliance with Regulations

The "Learn Mate" system is designed to comply with:

- Data protection regulations applicable to educational institutions
- Child online privacy protection requirements
- Institutional policies regarding student data management
- Future readiness for evolving data protection standards